



1 Decay Scheme

Bi-213 disintegrates 97.91 (3) % by beta minus emission to levels in Po-213 and 2.09 (3) % through alpha decay to Tl-209.

Le bismuth 213 se désintègre principalement par émissions bêta vers le polonium 213 et par émissions alpha (2,09 %) vers le thallium 209.

2 Nuclear Data

$T_{1/2}(^{213}\text{Bi})$	:	45,59	(6)	min
$T_{1/2}(^{213}\text{Po})$	:	3,70	(5)	10^{-6} s
$T_{1/2}(^{209}\text{Tl})$	:	2,161	(7)	min
$Q^{\alpha}(^{213}\text{Bi})$	:	5983	(6)	keV
$Q^{-}(^{213}\text{Bi})$	:	1423	(5)	keV

2.1 α Transitions

	Energy (keV)	Probability (%)	F
$\alpha_{0,1}$	5655 (10)	0,186 (5)	103
$\alpha_{0,0}$	5981 (10)	1,90 (4)	319

2.2 β^- Transitions

	Energy (keV)	Probability (%)	Nature	lg ft
$\beta_{0,9}^-$	95 (5)	0,0014 (2)		7,68
$\beta_{0,8}^-$	304 (5)	0,0608 (20)		7,07
$\beta_{0,7}^-$	323 (5)	0,595 (17)		6,16
$\beta_{0,6}^-$	377 (5)	0,020 (4)		7,85
$\beta_{0,5}^-$	419 (5)	0,0648 (23)		7,494
$\beta_{0,4}^-$	555 (5)	0,0129 (6)	Unique 1st Forbidden	8,597
$\beta_{0,3}^-$	822 (5)	0,0025 (19)		9,9
$\beta_{0,2}^-$	983 (5)	30,8 (4)	1st Forbidden	6,07
$\beta_{0,1}^-$	1130 (5)	0,21 (9)	1st Forbidden	8,45
$\beta_{0,0}^-$	1423 (5)	66,2 (4)	1st Forbidden	6,316

2.3 Gamma Transitions and Internal Conversion Coefficients

	Energy (keV)	P _{$\gamma+ce$} (%)	Multipolarity	α_K	α_L	α_M	α_N	α_O	α_T
$\gamma_{2,1}$ (Po)	147,70 (4)	0,0314 (20)	E2	0,307 (4)	0,85 (1)	0,226 (3)	0,0580 (8)	0,0110 (2)	1,453 (21)
$\gamma_{1,0}$ (Po)	292,80 (1)	0,55 (8)	M1+E2	0,22 (17)	0,06 (2)	0,015 (3)	0,0038 (7)	0,00077 (16)	0,30 (18)
$\gamma_{1,0}$ (Tl)	323,70 (2)	0,1866 (37)	M1+E2	0,134 (14)	0,0333 (13)	0,0081 (3)	0,00200 (7)		0,178 (15)
$\gamma_{5,3}$ (Po)	402,8 (3)	0,00010 (4)							
$\gamma_{2,0}$ (Po)	440,44 (1)	30,77 (36)	M1	0,146 (2)	0,0250 (4)	0,00594 (8)	0,00150 (2)	0,000320 (5)	0,179 (3)
$\gamma_{4,1}$ (Po)	574,9 (3)	0,00068 (16)							
$\gamma_{3,0}$ (Po)	600,9 (2)	0,0026 (19)							
$\gamma_{6,2}$ (Po)	604,98 (18)	0,0014 (5)							
$\gamma_{7,2}$ (Po)	659,759 (20)	0,043 (6)							
$\gamma_{5,1}$ (Po)	710,82 (3)	0,0112 (6)							
$\gamma_{7,1}$ (Po)	807,372 (11)	0,287 (14)							
$\gamma_{8,1}$ (Po)	826,564 (41)	0,0065 (4)							
$\gamma_{4,0}$ (Po)	867,961 (20)	0,0122 (6)							
$\gamma_{9,2}$ (Po)	887,76 (30)	0,00102 (19)							
$\gamma_{5,0}$ (Po)	1003,593 (17)	0,0535 (22)							
$\gamma_{6,0}$ (Po)	1045,68 (8)	0,019 (4)							
$\gamma_{7,0}$ (Po)	1100,17 (1)	0,265 (6)							
$\gamma_{8,0}$ (Po)	1119,422 (8)	0,0543 (20)							
$\gamma_{9,0}$ (Po)	1328,2 (3)	0,00039 (13)							

3 Atomic Data

3.1 Po

ω_K	:	0,965	(4)
$\bar{\omega}_L$	:	0,403	(16)
n_{KL}	:	0,807	(5)

3.1.1 X Radiations

	Energy (keV)		Relative probability
X _K			
	K α_2	76,864	60,05
	K α_1	79,293	100
	K β_3	89,256	} 34,43
	K β_1	89,807	
	K β_5''	90,363	
	K β_2	92,263	} 10,71
	K β_4	92,618	
	KO _{2,3}	92,983	
X _L			
	L ℓ	9,6576	
	L α	11,0161 - 11,1303	
	L η	12,0847	
	L β	12,8239 - 14,2476	
	L γ	15,251 - 16,2129	

3.1.2 Auger Electrons

		Energy (keV)	Relative probability
Auger K			
	KLL	58,978 - 65,205	100
	KLX	71,902 - 79,289	56,4
	KXY	84,8 - 93,1	7,95
Auger L		5,43 - 16,86	

3.2 Tl

ω_K

:

0,963

(4)

$\bar{\omega}_L$

:

0,367

(15)

n_{KL}

:

0,812

(5)

3.2.1 X Radiations

		Energy (keV)		Relative probability
X _K				
	Kα ₂	70,8325		59,24
	Kα ₁	72,8725		100
	Kβ ₃	82,118	}	34
	Kβ ₁	82,577		
	Kβ ₅ ''	83,115		
	Kβ ₂	84,838	}	10,1
	Kβ ₄	85,134		
	KO _{2,3}	85,444		
X _L				
	Lℓ	8,9531		
	Lα	10,1718 - 10,2679		
	Lη	10,9942		
	Lβ	11,8117 - 12,9566		
	Lγ	13,8528 - 14,7362		

3.2.2 Auger Electrons

		Energy (keV)	Relative probability
Auger K			
	KLL	54,587 - 59,954	100
	KLX	66,37 - 72,86	55,4
	KXY	78,12 - 85,50	7,67
Auger L			
		5,18 - 10,13	

4 α Emissions

	Energy (keV)	Probability (%)
$\alpha_{0,1}$	5549 (10)	0,186 (5)
$\alpha_{0,0}$	5869 (10)	1,90 (4)

5 Electron Emissions

		Energy (keV)	Electrons (per 100 disint.)
e _{AL}	(Po)	5,43 - 16,86	2,31 (4)
e _{AK}	(Po)		
	KLL	58,978 - 65,205	} 0,137 (16)
	KLX	71,902 - 79,289	
	KXY	84,8 - 93,1	
e _{AL}	(Tl)	5,18 - 10,13	0,0142 (6)
e _{AK}	(Tl)		
	KLL	54,587 - 59,954	} 0,00079 (12)
	KLX	66,37 - 72,86	
	KXY	78,12 - 85,50	
ec _{2,1} T	(Po)	54,6 - 147,7	0,0186 (12)
ec _{1,0} T	(Po)	199,7 - 292,8	0,13 (8)
ec _{1,0} T	(Tl)	238,2 - 323,7	0,0282 (24)
ec _{2,0} T	(Po)	347,30 - 440,41	4,67 (9)
ec _{2,0} K	(Po)	347,34 (1)	3,81 (7)
ec _{2,0} L	(Po)	423,51 - 426,63	0,653 (13)
ec _{2,0} M	(Po)	436,29 - 437,76	0,1550 (27)
$\beta_{0,9}^-$	max:	95 (5)	} 0,0014 (2)
	avg:	24,6 (14)	
$\beta_{0,8}^-$	max:	304 (5)	} 0,0608 (20)
	avg:	84,9 (16)	
$\beta_{0,7}^-$	max:	323 (5)	} 0,595 (17)
	avg:	90,8 (16)	
$\beta_{0,6}^-$	max:	377 (5)	} 0,020 (4)
	avg:	107,9 (16)	
$\beta_{0,5}^-$	max:	419 (5)	} 0,0648 (23)
	avg:	121,4 (17)	
$\beta_{0,4}^-$	max:	555 (5)	} 0,0129 (6)
	avg:	166,4 (17)	

		Energy (keV)		Electrons (per 100 disint.)	
$\beta_{0,3}^-$	max:	822	(5)	}	0,0025 (19)
	avg:	260,8	(19)		
$\beta_{0,2}^-$	max:	983	(5)	}	30,8 (4)
	avg:	320,4	(19)		
$\beta_{0,1}^-$	max:	1130	(5)	}	0,21 (9)
	avg:	376,8	(20)		
$\beta_{0,0}^-$	max:	1423	(5)	}	66,2 (4)
	avg:	492,2	(20)		

6 Photon Emissions

6.1 X-Ray Emissions

		Energy (keV)	Photons (per 100 disint.)		
XL	(Po)	9,6576 - 16,2129	1,54 (4)		
XK α_2	(Po)	76,864	1,103 (29)	}	K α
XK α_1	(Po)	79,293	1,84 (5)		
XK β_3	(Po)	89,256	0,633 (19)	}	K' β_1
XK β_1	(Po)	89,807			
XK β_5''	(Po)	90,363			
XK β_2	(Po)	92,263	0,197 (7)	}	K' β_2
XK β_4	(Po)	92,618			
XKO _{2,3}	(Po)	92,983			
XL	(Tl)	8,9531 - 14,7362	0,0083 (5)		
XK α_2	(Tl)	70,8325	0,0060 (7)	}	K α
XK α_1	(Tl)	72,8725	0,0101 (11)		
XK β_3	(Tl)	82,118	0,0034 (4)	}	K' β_1
XK β_1	(Tl)	82,577			
XK β_5''	(Tl)	83,115			
XK β_2	(Tl)	84,838	0,00102 (11)	}	K' β_2
XK β_4	(Tl)	85,134			
XKO _{2,3}	(Tl)	85,444			

6.2 Gamma Emissions

	Energy (keV)	Photons (per 100 disint.)
$\gamma_{2,1}(\text{Po})$	147,70 (4)	0,0128 (8)
$\gamma_{1,0}(\text{Po})$	292,80 (1)	0,421 (7)
$\gamma_{1,0}(\text{Tl})$	323,70 (2)	0,1584 (24)
$\gamma_{5,3}(\text{Po})$	402,8 (3)	0,00010 (4)
$\gamma_{2,0}(\text{Po})$	440,44 (1)	26,1 (3)
$\gamma_{4,1}(\text{Po})$	574,9 (3)	0,00068 (16)
$\gamma_{3,0}(\text{Po})$	600,9 (2)	0,0026 (19)
$\gamma_{6,2}(\text{Po})$	604,93 (17)	0,0014 (5)
$\gamma_{7,2}(\text{Po})$	659,75 (2)	0,043 (6)
$\gamma_{5,1}(\text{Po})$	710,82 (3)	0,0112 (6)
$\gamma_{7,1}(\text{Po})$	807,37 (1)	0,287 (14)
$\gamma_{8,1}(\text{Po})$	826,55 (4)	0,0065 (4)
$\gamma_{4,0}(\text{Po})$	867,96 (2)	0,0122 (6)
$\gamma_{9,2}(\text{Po})$	886,66 (14)	0,00102 (19)
$\gamma_{5,0}(\text{Po})$	1003,58 (2)	0,0535 (22)
$\gamma_{6,0}(\text{Po})$	1045,67 (8)	0,019 (4)
$\gamma_{7,0}(\text{Po})$	1100,16 (1)	0,265 (6)
$\gamma_{8,0}(\text{Po})$	1119,42 (8)	0,0543 (20)
$\gamma_{9,0}(\text{Po})$	1328,2 (3)	0,00039 (13)

7 Main Production Modes

Np – 237 decay chain

Th – 229 decay chain

Ac – 225 decay chain

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